

Comparing Theories that Posit a Role for Task Features in Strategy Selection

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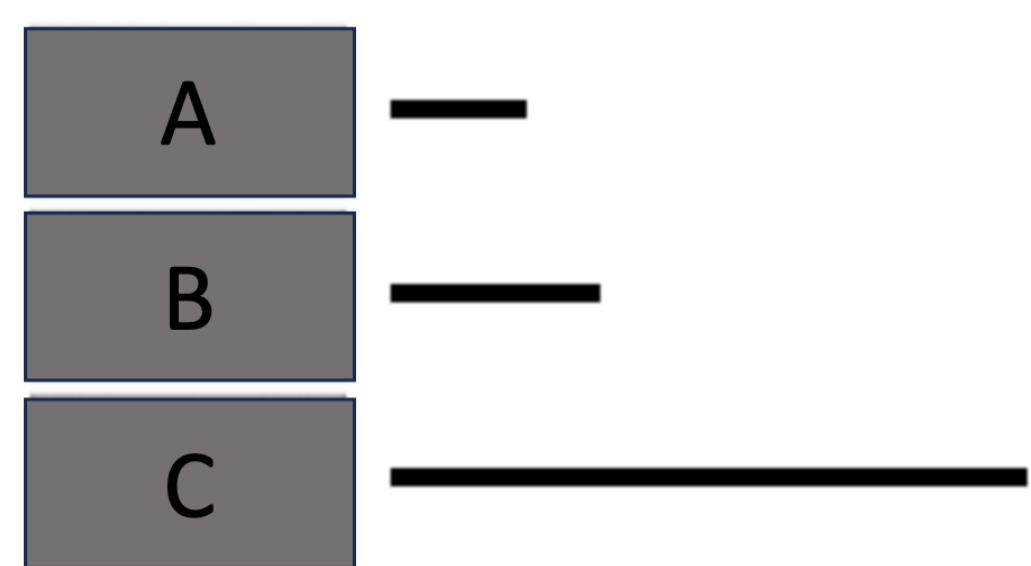
Introduction

- Salient features of a task play an important role in how people create task representations which then influence strategy selection for accomplishing the task. We examined two theories, Represent-Construct-Choose-Learn (RCCL; Lovett & Schunn, 1999) and Rational Metareasoning (RM; Lieder & Griffiths, 2007), both of which incorporate task features into their models of strategy selection.
- RCCL: when a strategy’s success rate is low, it indicates that the task representation is not useful and those represented features are irrelevant in this case, so people tend to **drop these features from the task representation**.
- RM: strategy selection is based on consideration of all available features, with **no discrete changes in the features incorporated into the task representation**.
- Goal:** By manipulating features in a simple problem-solving task, our study attempts to determine if unhelpful features are dropped from the problem representation as explained by RCCL or will continue to be used to select strategies as explained by RM.

Methods

- Participants:** N = 259
- Task:** Building Stick Task (BST; Schunn et al., 2001)
 - The goal of the task is to add and subtract the lengths of three sticks to match the target length.
 - A participant's strategy can be categorized as either undershoot or overshoot based on the first move.

Undershoot



Overshoot

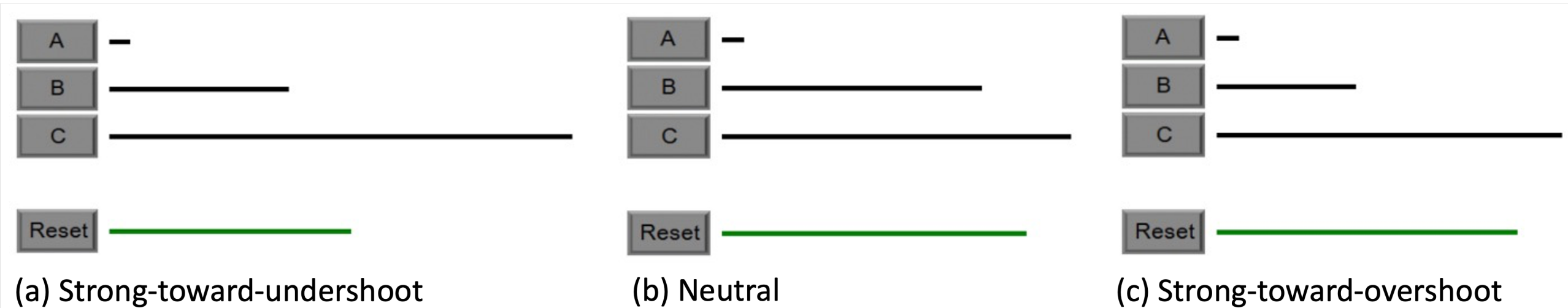


Target Stick

B+A+A

C-A-A

Three types of problems:



- Design:**
 - 2 (correct strategy during the experimental phase: undershoot vs. overshoot) * 4 (problem-group: cue-inconsistent, no-cue, cue-consistent, mixed) between-participant design

Procedure:

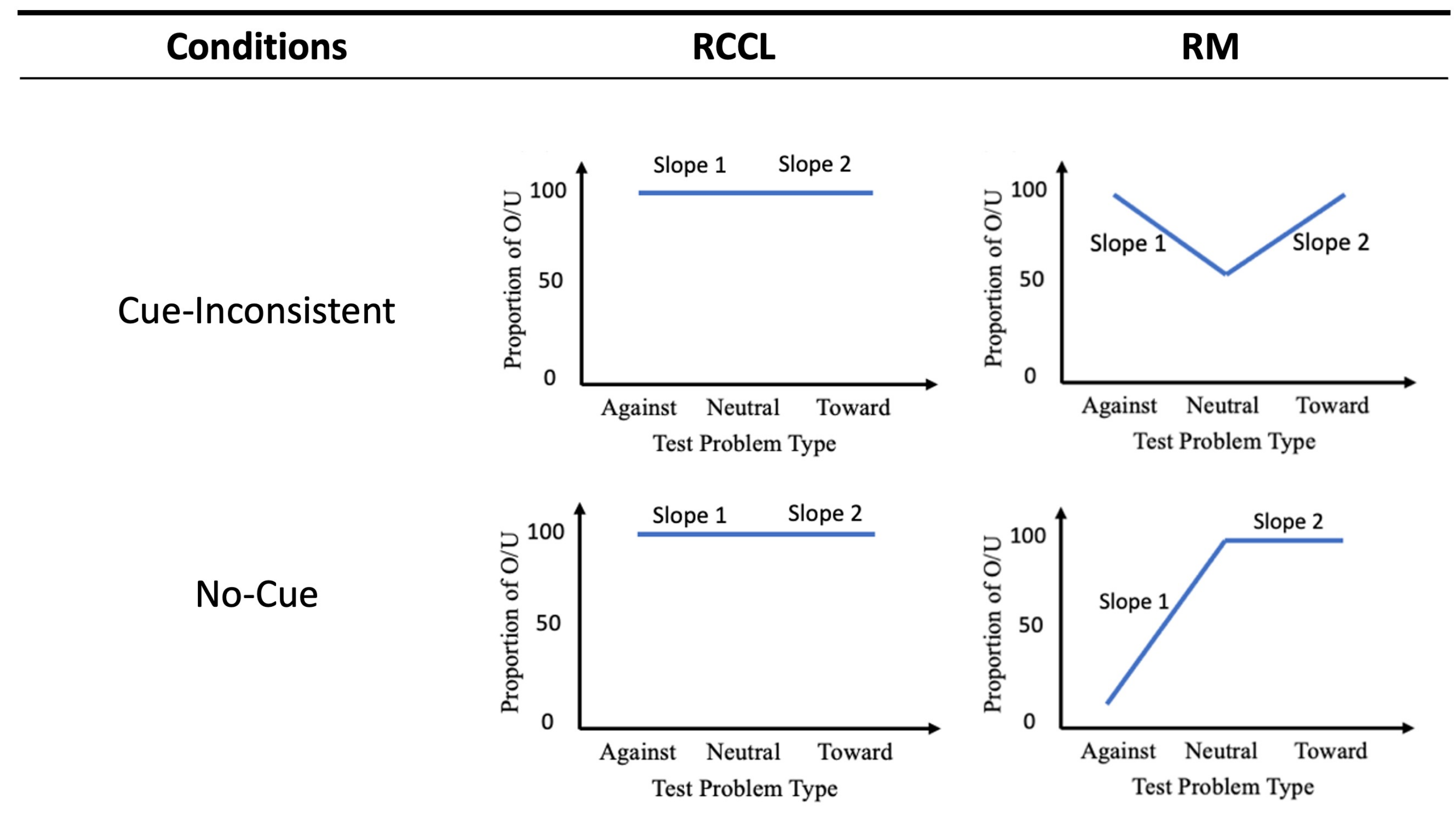
Relative-length-cue training

Pretest

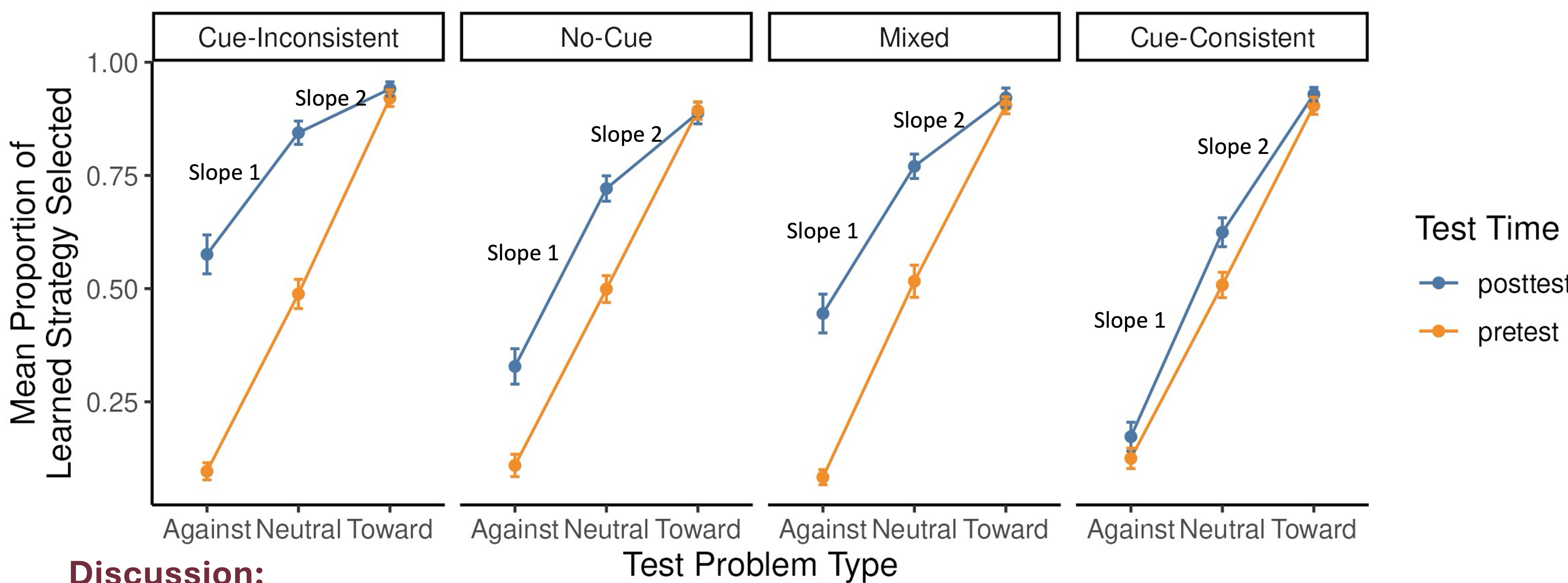
Experimental

Posttest

Predictions:



Results and Discussion



Discussion:

- Why is there a big difference between our prediction and data? To address this question, we revisited the predictions.
- Notably, in RM, the relative length cue pointing to the overshoot strategy was regarded as a separate feature from the feature indicting the length cue pointing to the undershoot strategy. However, if the relative length feature is viewed as continuous, the prediction should be different from the earlier prediction that the values tied with overshoot and undershoot vary separately. Under the newer prediction, the strategy selection will change in a manner more closely aligned with the data.
- In conclusion, our data showed evidence against the RCCL theory, but it may support the RM theory when different ways to define a feature are considered.

For the posttest:

- In cue-inconsistent and no-cue conditions, our data did not fully align either the RCCL or RM.
- In the mixed condition, a significant slope change from pretest to posttest provided support for the RM theory.
- Cue-consistent served as a control condition.

Acknowledgement & References

This work was supported by Office of Naval Research grant N00014-21-1-2617.

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